

BLUE REPORT

Routine primary tumor resection does not improve overall survival in de novo metastatic HR+/HER2-negative breast cancer and is **not recommended as standard practice** according to current evidence.[\[1\]\[2\]](#) However, locoregional surgery **may be considered in highly selected patients** with favorable features, particularly those with oligometastatic disease.[\[2\]](#)

Evidence from Randomized Controlled Trials

The most recent Cochrane systematic review (2025) including 5 randomized controlled trials with 1,368 women found that breast surgery does not improve overall survival in de novo metastatic breast cancer (HR 0.89, 95% CI 0.75-1.05).[\[1\]](#) While exploratory subgroup analyses suggested a potential survival benefit in women with luminal (HR+) tumors (HR 0.82, 95% CI 0.69-0.96), these findings are not definitive.[\[1\]](#) Surgery consistently improved local disease control (HR 0.43, 95% CI 0.32-0.58) but did not improve distant progression-free survival or quality of life.[\[1\]](#)

A 2024 meta-analysis of 6 RCTs found a modest overall survival benefit with locoregional surgery (HR 0.86, 95% CI 0.77-0.96), with more pronounced effects in hormone receptor-positive patients (HR 0.79, $p=0.003$) and HER2-negative patients (HR 0.80, $p=0.003$).[\[3\]](#) However, another 2024 meta-analysis found no overall survival benefit (HR 0.93, 95% CI 0.76-1.14) except possibly in younger/premenopausal patients (HR 0.74, 95% CI 0.58-0.94).[\[4\]](#)

The JCOG1017 trial (2025) randomized 407 patients who had not progressed after 3 months of systemic therapy and found no significant overall survival difference (HR 0.857, 90% CI 0.686-1.072), though surgery improved local relapse-free survival (median 63 vs 20 months).[\[5\]](#) Subgroup analyses suggested potential benefit in premenopausal patients and those with single-organ metastasis.[\[5\]](#)

NCCN Guidelines Recommendations

The NCCN Breast Cancer Guidelines (2026) state that "routine utilization of locoregional treatment of the primary tumor and ablative treatment for metastatic sites have not been demonstrated to improve survival outcomes in prospective trials." However, the guidelines acknowledge that for[\[2\]](#) **de novo oligometastatic disease with favorable features** (HER2+ or ER+, induced no evidence of disease status after optimal systemic therapy, limited sites of metastases, bone-only metastases, and young age/longevity), "an aggressive multimodality approach involving locoregional therapy of the primary tumor, ablative treatment to all sites of metastases, and 'adjuvant-like' systemic therapy may offer improved long-term progression-free survival and may be considered."[\[2\]](#)

Circumstances Where Surgery May Be Considered

Based on current evidence, locoregional surgery might be considered in this patient if she has:

Favorable prognostic features present in this case:

- Young age (56 years, postmenopausal but relatively young)
- Excellent performance status (healthy and active)
- Limited metastatic burden (ovaries only, no other sites on PET/CT)
- HR+/HER2-negative disease
- Ability to achieve complete resection with negative margins

Additional considerations:

- Response to initial systemic therapy should be documented before considering surgery[\[2\]\[5\]](#)
- Surgery should only be pursued after multidisciplinary discussion[\[2\]\[6\]](#)
- The primary benefit would be improved local control, not necessarily survival[\[1\]\[5\]](#)
- Patient motivation and preferences are important, but should be balanced against evidence showing no clear survival benefit in most patients[\[2\]](#)

Recommended Approach

For this patient, the evidence-based approach would be:

1. **Initiate systemic therapy** as standard of care (endocrine therapy plus CDK4/6 inhibitor given HR+/HER2-negative disease)[\[2\]](#)
2. **Reassess after 3-6 months** of systemic therapy to document response
3. **Multidisciplinary discussion** regarding potential surgery if she has excellent response and remains motivated
4. If surgery is pursued, ensure **complete resection with negative margins** and plan for comprehensive locoregional therapy[\[2\]\[5\]](#)

5. Continue systemic therapy as the primary treatment modality[2]

The patient should be counseled that while surgery may improve local control and is reasonable to consider given her favorable features, current high-quality evidence does not demonstrate a clear overall survival benefit for routine primary tumor resection in de novo metastatic disease.[1][7][4]

Would you like me to explore the specific systemic therapy options for HR+/HER2-low metastatic breast cancer with PIK3CA mutation, given the molecular profiling results?

References

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2. [Breast Cancer](#). National Comprehensive Cancer Network. Updated 2026-01-16.
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4. [Impact of Primary Breast Surgery on Overall Survival of Patients With De Novo Metastatic Breast Cancer: A Systematic Review and Meta-Analysis](#). Villacampa G, Papakonstantinou A, Fredriksson I, Matikas A. The Oncologist. 2024;29(1):1-7. doi:10.1093/oncolo/oyad266.
5. [Primary Tumour Resection Plus Systemic Therapy Versus Systemic Therapy Alone in Metastatic Breast Cancer \(JCOG1017, PRIM-BC\): A Randomised Clinical Trial](#). Shien T, Hara F, Aogi K, et al. British Journal of Cancer. 2025;:10.1038/s41416-025-03097-z. doi:10.1038/s41416-025-03097-z.
6. [Locoregional Therapy of the Primary Tumour in De Novo Stage IV Breast Cancer in 216 066 Patients: A Meta-Analysis](#). Gera R, Chehade HELH, Wazir U, et al. Scientific Reports. 2020;10(1):2952. doi:10.1038/s41598-020-59908-1.
7. [Primary-Site Local Therapy for Patients With De Novo Metastatic Breast Cancer: An Educational Review](#). Khan SA, Schuetz S, Hosseini O. Annals of Surgical Oncology. 2022;29(9):5811-5820. doi:10.1245/s10434-022-11900-x.